

Credit Card Approval Prediction Using Machine Learning

Abstract

Banks receive thousands of credit card applications every day. This project automates approval prediction using Machine Learning and a Flask web application.

Introduction

The system predicts whether an applicant is likely to be approved or rejected based on historical applicant data.

Problem Statement

Manual credit card screening is time-consuming and prone to human error.

Objectives

Automate approval prediction, compare ML models, and provide a Flask dashboard.

Dataset

application_record.csv and credit_record.csv were merged using the ID column.

Preprocessing

Cleaning, handling missing values, label encoding, feature selection, and train-test split.

Algorithms

Logistic Regression, Decision Tree, Random Forest, and XGBoost.

Results

Random Forest achieved the best accuracy of 98.69%.

Technologies

Python, Flask, Pandas, NumPy, Scikit-learn, XGBoost, HTML, CSS, Bootstrap.

Future Scope

Cloud deployment, authentication, database integration, and model improvement.

Conclusion

The project demonstrates an automated approach to credit card approval prediction.